Exp :05

Write the python program for Missionaries Cannibal problem.

Aim:

To write a python program for Missionaries Cannibal problem.

Objective:

To understand state-space search with constraints by transporting 3 missionaries and 3 cannibals across a river using a boat, without cannibals ever outnumbering missionaries on either bank.

Algorithm:

Step 1: Create a data structure to represent the state of the problem, including the number of missionaries and cannibals on each side of the river and the position of the boat.

Step 2: Create a data structure to represent the state space of the problem, including the initial state, the goal state, and the possible transitions between states.

Step 3: Use a search algorithm, such as depth-first search or breadth-first search, to explore the state space and find a solution.

Step 4: At each step of the search, generate all possible successor states from the current state by applying one of the allowed boat movements (two missionaries, two cannibals, or one of each).

Step 5: Check if the successor state is valid (i.e., no more cannibals than missionaries on either side of the river).

Step 6: If the successor state is valid, add it to the search tree and continue the search.

Step 7: If the successor state is the goal state, return the path from the initial state to the goal state.

Step 8: If there are no more possible successor states to explore, backtrack to the previous state and try another possible movement.

Program:

```
left_m = 3
left_c = 3
right_m = 0
```

```python
right_c = 0
boat = "left"

print("Game Start\n")
print("Now the task is to move all of them to right side of the river")
print("Rules:")
print("1. The boat can carry at most two people")
print("2. If cannibals are greater than missionaries, the missionaries will be eaten")
print("3. The boat cannot cross the river by itself with no people on board\n")

while True:

    print("\nLeft Bank : ", "M " * left_m + "C " * left_c)
    print("Right Bank:", "M " * right_m + "C " * right_c)

    if left_m == 0 and left_c == 0:
        print("\nCongratulations! You solved the problem.")
        break

    print("\n", boat.capitalize(), "side ->", "Right" if boat == "left" else "Left", "side
river travel")

    m = int(input("Enter number of Missionaries travel => "))
    c = int(input("Enter number of Cannibals travel => "))

    if m + c == 0 or m + c > 2:
        print("Invalid input please retry !!")
        continue

    if boat == "left":
        if m > left_m or c > left_c:
            print("Invalid input please retry !!")
            continue
```

```
        left_m -= m
        left_c -= c
        right_m += m
        right_c += c
        boat = "right"

    else:
        if m > right_m or c > right_c:
            print("Invalid input please retry !!")
            continue

        right_m -= m
        right_c -= c
        left_m += m
        left_c += c
        boat = "left"

    if (left_m > 0 and left_c > left_m) or (right_m > 0 and right_c > right_m):
        print("\nCannibals ate the Missionaries!")
        print("Game Over.")
        break
```

Output:

Game Start

Now the task is to move all of them to right side of the river

Rules:

1. The boat can carry at most two people
2. If cannibals are greater than missionaries, the missionaries will be eaten
3. The boat cannot cross the river by itself with no people on board

Left Bank : M M M C C C

Right Bank:

Left side -> Right side river travel

Enter number of Missionaries travel =>

Result:

The program was executed and the output was found to be correct